Department of Electrical and Electronics Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE 'A'
Course Code & Title: EE8552 Power Electronics
Name of the Faculty member: Mr. S. Meenakshi Sundaravel

Innovative Practice Description

- **Unit / Topic:** Unit II / Phase Controlled Converters
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 5
- **Activity Chosen:** Active Learning Method – MATLAB Simulation
- **Justification:**

Rectifiers can be classified as single phase rectifier and three phase rectifier. Single phase rectifiers are classified as 1-Φ half wave and 1-Φ full wave rectifier. Three phase rectifier are classified as 3-Φ half wave rectifier and 3-Φ full wave rectifier. 1-Φ Full wave rectifiers are classified as1-Φmid point type and 1-Φ bridge type rectifier. 1-Φ bridge type rectifiers are classified as 1-Φ half controlled and 1-Φ full controlled rectifier. 3-Φ full wave rectifiers are again classified as 3-Φmid point type and 3-Φ bridge type rectifier. 3-Φ bridge type rectifiers are again divided as 3-Φ half controlled rectifier and 3-Φ full controlled rectifier. After teaching 1-Φ full wave rectifier and 3-Φ full wave rectifier, I had conducted a simulation activity for making the students to understand the operation of both in practical mode and also enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

After teaching the concept, gave students five to ten minutes to think about the topicwithout writing anything.

Total Strength is 31.

Photographer: A student – Mr. S. Arunkumar (953620105304)

Reporter: Myself

At the end the Class (Last 10 minutes)

- ✓ I asked the students to simulate the 1-Φ full wave rectifier and 3-Φ full wave rectifier for 10 minutes.

- ✓ Then I asked them to obtain the pulsating output voltage waveform of them for another 5 minutes.
- ✓ Finally, I randomly asked students about the operation of 1-Φ full wave rectifier and 3-Φ full wave rectifier for 5 minutes.

- **CO – PO / PSO mapping:**

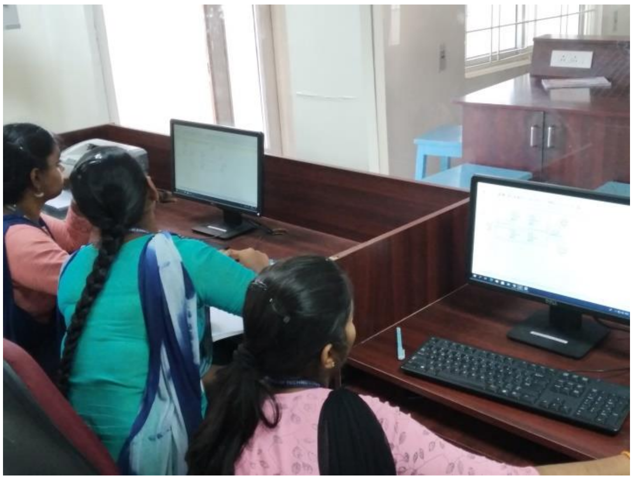
CO	PO 1	PO 2	PO 3	PO 5	PO10	PO12	PSO1	PSO3
CO 2	3	2	1	3	1	1	2	1

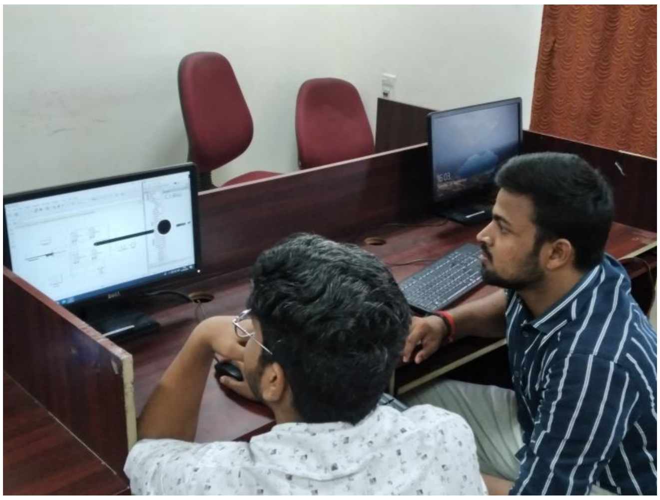
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 5** **3**
Justification for correlation	Students have utilized the modern tool MATLAB for performing simulation. Hence, this activity is correlated with level 3.

- **Images / Screenshot of the practice:**





Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

  Students were able to perform the simulation of 1-Φ full wave rectifier and 3-Φ full wave rectifier and also they said that they understood the functions of them.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society.
2. Students understood the concept which was reflected from their answers for the questions I have asked.
3. The success of the activity was evaluated by asking the same question during IAT – I **– Around 80% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity, slow learners were not able to perform the simulation quickly.

References:

✓ P.S.Bimbra "Power Electronics" Khanna Publishers, Third Edition, 2003.

Signature of Faculty Member **HOD**

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Course Code & Title: EE8552 Power Electronics

Name of the Faculty member: Mr. S. Meenakshi Sundaravel

Innovative Practice Description

- **Unit / Topic:** Unit IV / Inverters
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 16
- **Activity Chosen:** Active Learning Method – Demonstration
- **Justification:**

Inverters can be classified as single phase inverter and three phase inverter. After teaching 1-Φ inverter and 3-Φ inverter, I had conducted this demonstration activity for making the students to understand the operation of inverters both in theoretical and practical mode and also enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**

After teaching the concept, gave students five to ten minutes to think about the topicwithout writing anything.

Total Strength is 31.

Photographer: A student – Mr. S. Arunkumar (953620105304)

Reporter: Myself

At the end the Class (Last 10 minutes)

- ✓ I asked the students to explain the operation of 1-Φ and 3-Φ inverter for 10 minutes.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO10	PO12	PSO1	PSO3
CO 2	3	2	1	2	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 2**
	2
Justification for correlation	Due to the demonstration of inverter, students can able to identify, formulate and analyze the complex engineering problem statements.

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  Students were able to explain the operation of 1-Φ and 3-Φ inverters and also they said that they understood the functions of them.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society.
2. Students understood the concept which was reflected from their answers for the questions I have asked.
3. The success of the activity was evaluated by asking the same question during IAT – III **– Around 80% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

✓ P.S.Bimbra "Power Electronics" Khanna Publishers, Third Edition, 2003.

Signature of Faculty Member **HOD**